

Dhananjay R

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Education

Manipal School of Information Sciences, MAHE <i>Master of Engineering in Big Data Analytics (CGPA: 8.04)</i>	Manipal, India July 2025 – Present
Acharya Institute of Technology <i>Bachelor of Engineering in Computer Science and Engineering (CGPA: 7.02)</i>	Bengaluru, India Dec 2020 – May 2023

Skills Summary

Languages: Python, Java, SQL
Data Analysis & Manipulation: MySQL, NumPy, Pandas, PySpark, Power BI, Scrapy, BeautifulSoup
Data Engineering: Hadoop, Spark, Hive, Sqoop, Apache Airflow, Docker, MinIO, Snowflake
ML / Deep Learning: LangChain, HuggingFace, FAISS, Pinecone, Transformers, TensorFlow, Scikit-learn
Backend & APIs: Flask, REST APIs, PostgreSQL
Cloud & DevOps: AWS (Cloud Foundations), Oracle Cloud Infrastructure, Linux (Bash, CLI), Git, CI/CD

Experience

Spherenex Innovation Labs <i>Software Engineer</i>	Bengaluru, Karnataka Sept 2023 – Nov 2024
- Translated HLL wireframes into production-ready frontends using Angular, Ionic, and TypeScript, following component-driven and scalable architecture patterns. - Integrated RESTful APIs across frontend modules, managing async data flows and validating endpoints end-to-end using Postman for reliable client-server communication. - Executed Functional Testing and UAT cycles – authoring test cases, triaging defects, and ensuring stakeholder-aligned releases with zero critical regression. - Delivered sprint increments within Agile workflows, maintaining code quality and modularity through Git version control and CI/CD pipeline practices.	

MCA Coordinator <i>Vidyavahini First Grade College</i>	Tumkur, Karnataka May 2025 – July 2025
- Led MCA department training and placement operations, delivering advanced technical sessions on Java, Spring Boot, Angular.js, and REST APIs aligned with current industry demands. - Mentored students on capstone projects, resume building, and interview preparation, while collaborating with industry partners for campus recruitment and internship drives. - Designed placement-oriented workshops and skill development initiatives to bridge the gap between academic curriculum and industry expectations.	

Lecturer <i>Aryabharathi Polytechnic</i>	Tumkur, Karnataka Jan 2025 – May 2025
- Delivered structured lectures on Java and Computer Science fundamentals to diploma students, simplifying complex concepts for non-technical learners. - Designed course content and mentored students for industry readiness, academic progression, and foundational programming skills.	

Projects

OPD-RAG	2026
- Developed OPD-RAG, a Retrieval-Augmented Generation (RAG) application for semantic question answering over OPD healthcare records, leveraging LangChain for orchestration, FAISS for vector indexing, and cosine similarity for efficient patient data retrieval. - Integrated the LLaMA 7B model to generate accurate, context-aware clinical responses across 1000+ OPD records, delivering an end-to-end intelligent healthcare querying solution.	

- Engineered a multimodal RAG pipeline integrating **SentenceTransformers (MiniLM)** and **FAISS** to perform sub-second semantic retrieval over **168K+ vector embeddings** derived from 50K+ product records.
- Developed an image-to-text retrieval system using **BLIP (image captioning)** with preprocessing and heuristic filtering to convert visual inputs into optimized semantic queries for vector search.
- Built an end-to-end LLM-powered retrieval system using **Flask REST APIs** and **LLaMA (HuggingFace)**, enabling context-aware product recommendations through dynamic prompt construction and top-K similarity search.

Wikipedia Trending Topic Analysis

2026

- Developed a Wikipedia Trending Topics Analysis data engineering project by ingesting data from the **Wikimedia Pageview API** and building an automated ETL pipeline using **Apache Airflow** and **PySpark**.
- Containerized the workflow with **Docker**, stored processed data in **MinIO**, and created interactive analytical dashboards using **Metabase** for trend monitoring and insights.

Demand-Supply Chain Management System – IEEE Published

2023

ML-Based Crop Demand Optimization

- Built predictive models using **XGBoost** for crop yield estimation and demand forecasting, improving decision-making for optimal crop selection.
- Designed a recommendation system to suggest alternative crops based on soil, climate, and market demand, enhancing supply-demand balance.
- Developed RESTful APIs using **FastAPI** integrated with **PostgreSQL** for efficient data storage, retrieval, and real-time prediction serving.
- Conducted model evaluation and feature engineering to improve prediction accuracy, and published the research findings in **IEEE**.

Certifications and Achievements

- **Generative AI with LLMs** — Coursera (2025)
- **AWS Academy Graduate** — **Cloud Foundations** (2025)
- **Oracle Cloud Infrastructure** — **Certified Data Science Professional** (2025)
- **Snowflake** — Fundamentals, Data Engineering, Advanced Analytics (2025)
- Conducted **3–4 workshops** on **Java Full Stack Development**, delivering 1-week intensive training (5–6 hrs/day) with hands-on project implementation.